

Topics in Error-Correcting Codes

MIT 6.S953 — Problem Set 1

Submission Instructions: Problem 0 is not for submission. Choose any three of Problems 1–5 to solve by 6 PM, October 7. A link for scheduling a meeting with Rohan to present will be updated here soon.

Throughout, $\Delta(u, v)$ denotes Hamming distance and $\text{wt}(u)$ denotes Hamming weight. For a linear code, $(n, k, d)_q$ denotes its block length, dimension, minimum distance, and alphabet size.

Problem 0: Prime fields and Bézout’s identity

Let p be a prime and let $\mathbb{F}_p = \{0, 1, \dots, p-1\}$. To add or multiply two elements of $\mathbb{F}_p$, take their ordinary integer sum or product and then take the remainder upon division by p . We will show that $\mathbb{F}_p$ with these operations is a field.

- (a) Prove Bézout’s identity for coprime integers: if $a, b \in \mathbb{Z}$ satisfy $\gcd(a, b) = 1$, then there exist $u, v \in \mathbb{Z}$ such that

$$ua + vb = 1.$$

Hint: Apply the Euclidean algorithm, or consider the least positive integer of the form $ua + vb$.

- (b) Recall the field axioms below. Here $+$ and multiplication denote the operations on remainders defined above, and the identities must hold for every $x, y, z \in \mathbb{F}_p$.

- **Closure:** $x + y \in \mathbb{F}_p$ and $xy \in \mathbb{F}_p$.
- **Associativity:** $(x + y) + z = x + (y + z)$ and $(xy)z = x(yz)$.
- **Commutativity:** $x + y = y + x$ and $xy = yx$.
- **Identities:** The elements 0 and 1 are distinct and satisfy $x + 0 = x$ and $x1 = x$.
- **Additive inverses:** For every $x \in \mathbb{F}_p$, there exists $y \in \mathbb{F}_p$ such that $x + y = 0$.
- **Multiplicative inverses:** For every $x \in \mathbb{F}_p \setminus \{0\}$, there exists $y \in \mathbb{F}_p$ such that $xy = 1$.
- **Distributivity:** $x(y + z) = xy + xz$.

The interesting axiom is the existence of multiplicative inverses; the other axioms are straightforward to check. Prove only this axiom: for every $a \in \{1, \dots, p-1\}$, show that there exists $b \in \{1, \dots, p-1\}$ such that the remainder of the integer product ab upon division by p is 1.

- (c) Prove that $\{0, 1, \dots, m-1\}$, with addition and multiplication followed by taking the remainder upon division by m , is not a field when $m > 1$ is composite.

Problem 1: The Hamming code hat puzzle

Restatement of Guruswami–Rudra–Sudan, Essential Coding Theory, Chapter 1, Exercise 1.18, pp. 24–25 (April 19, 2026 draft). [Original exercise.](#)

There are $n \geq 1$ players in a room. Each player is given either a black hat or a white hat, with equal probability and independently of the other players. A player can see everyone else’s hat, but cannot see their own.

The players may agree on a strategy before the hats are assigned. After that, they cannot communicate. Each player must either guess the color of their own hat or abstain. All players make their decisions simultaneously, so nobody knows anyone else's decision when choosing their own.

The players win or lose as a team. They win if at least one player makes a guess and every player who guesses is correct. They lose if everyone abstains or if even one guess is incorrect. The goal is to find a strategy that maximizes the probability of winning.

- (a) **A warmup.** Describe a strategy that wins with probability at least $1/2$, for every number of players n .
- (b) **A graph formulation.** Represent a hat assignment by a string in $\{0, 1\}^n$, with 0 for black and 1 for white. The n -dimensional hypercube is the graph whose vertices are these strings, with an edge between two strings exactly when they differ in one coordinate.

Form a directed graph G as follows: keep all the vertices, and for each edge of the hypercube, either omit the edge or choose one of its two directions. Let $K(G)$ be the number of vertices of G with at least one incoming edge and no outgoing edges.

Relate the players' strategies to these directed graphs, and show that the largest possible winning probability is

$$\max_G \frac{K(G)}{2^n}.$$

Explain which vertices represent winning hat assignments.

- (c) **An upper bound.** Each vertex of G has at most n outgoing edges. Use this fact to prove that every such graph satisfies

$$\frac{K(G)}{2^n} \leq \frac{n}{n+1}.$$

Conclude that no strategy for the hat game wins with probability greater than $n/(n+1)$.

- (d) **Achieving the bound.** Suppose $n = 2^r - 1$ for an integer $r \geq 2$. Use the length- n Hamming code to construct a directed graph G satisfying $K(G)/2^n = n/(n+1)$, and hence obtain a strategy attaining the upper bound in part (c). You may use the parameters of Hamming codes from class. If you have only seen the length-7 Hamming code, you may do this part for $n = 7$.

Problem 2: The q -ary Johnson bound

In class, we showed that for $J(\delta) = 1 - \sqrt{1 - \delta}$, a code $\mathcal{C}$ of relative distance δ is $(J(\delta) - \varepsilon, 1/\varepsilon^2)$ list-decodable for every $0 < \varepsilon < J(\delta)$.

Show that this bound can be strengthened using the alphabet size. For an integer $q \geq 2$ and $0 < \delta \leq 1 - 1/q$, define

$$J_q(\delta) = \left(1 - \frac{1}{q}\right) \left(1 - \sqrt{1 - \frac{q\delta}{q-1}}\right).$$

Prove that every code $\mathcal{C} \subseteq \Sigma^n$ with $|\Sigma| = q$ and relative distance at least δ is $(J_q(\delta) - \varepsilon, 1/\varepsilon^2)$ list-decodable for every $0 < \varepsilon < J_q(\delta)$.

Recall the usual proof. Fix L distinct codewords within distance ρn of a received word y , and let a_i count those agreeing with y_i . Each codeword agrees with y in at least $(1 - \rho)n$ coordinates, so $\sum_i a_i \geq L(1 - \rho)n$.

Let A count agreements over all coordinates and ordered pairs of these codewords, including pairs of a codeword with itself. The L self-pairs contribute Ln agreements, while each pair of distinct codewords agrees in at most $(1 - \delta)n$ coordinates. Thus

$$A \leq Ln + L(L - 1)(1 - \delta)n.$$

At coordinate i , the a_i codewords agreeing with y_i also agree with one another, contributing a_i^2 ordered agreements. Counting just these pairs and applying Cauchy–Schwarz gives

$$A \geq \sum_{i=1}^n a_i^2 \geq \frac{(\sum_i a_i)^2}{n} \geq L^2(1 - \rho)^2 n.$$

Comparing these bounds gives the usual Johnson radius.

Hint: The lower bound ignores pairs that agree with each other on a symbol different from y_i . At coordinate i , the $L - a_i$ remaining codewords use only $q - 1$ other symbols. In the binary case, all of them must agree on the other symbol. For general q , use Cauchy–Schwarz to recover this missing contribution: look for an additional term involving $(L - a_i)^2/(q - 1)$. Combine the improved lower bound with the same upper bound, and set $\rho = J_q(\delta) - \varepsilon$ to bound L .

Problem 3: Hadamard codes and duality with Hamming codes

Let $r \geq 2$ and $n = 2^r - 1$. Index coordinates by the nonzero vectors $x \in \mathbb{F}_2^r$. Define

$$\text{Had}_r = \{(a \cdot x)_{x \in \mathbb{F}_2^r \setminus \{0\}} : a \in \mathbb{F}_2^r\} \subseteq \mathbb{F}_2^n,$$

where $a \cdot x = \sum_{j=1}^r a_j x_j$ is computed over $\mathbb{F}_2$. This is the punctured Hadamard code: the usual length- 2^r version also includes the coordinate $x = 0$, whose value is always zero.

For a linear code $\mathcal{C} \subseteq \mathbb{F}_q^N$, its *dual code* is

$$\mathcal{C}^\perp = \{z \in \mathbb{F}_q^N : z \cdot c = 0 \text{ for every } c \in \mathcal{C}\}.$$

We use the lecture convention that codewords are row vectors: a generator matrix is $G \in \mathbb{F}_q^{k \times N}$ and a parity-check matrix is $H \in \mathbb{F}_q^{N \times (N-k)}$, with $\mathcal{C} = \{c : cH = 0\}$.

- (a) **General duality.** Let $\mathcal{C} \subseteq \mathbb{F}_q^N$ have dimension k , generator matrix G , and parity-check matrix H . Prove that

$$\dim(\mathcal{C}^\perp) = N - k, \quad (\mathcal{C}^\perp)^\perp = \mathcal{C}.$$

Show that H^T is a generator matrix for $\mathcal{C}^\perp$ and G^T is a parity-check matrix for $\mathcal{C}^\perp$.

- (b) **Hadamard parameters.** Prove that Had_r is linear and that its encoding map is injective. Show that every nonzero codeword has weight exactly 2^{r-1} , and conclude that its parameters are

$$(2^r - 1, r, 2^{r-1})_2.$$

What are the parameters if the coordinate $x = 0$ is included?

- (c) **Parameters of the dual.** Prove that $\text{Had}_r^\perp$ is a binary linear code with parameters

$$(2^r - 1, 2^r - r - 1, 3)_2.$$

Observe that these are the same as the parameters of Ham_r .

Remark. Ham_r is in fact the dual of Had_r , up to reordering coordinates!

Problem 4: Characteristic and constructing finite fields

The *characteristic* of a field K , denoted $\text{char}(K)$, is the least positive integer t such that $t \cdot 1_K = 0$, where $t \cdot 1_K$ is the sum of t copies of 1_K . If no such integer exists, the characteristic is 0.

- (a) Show that the characteristic of any field is either 0 or a prime. Show that every finite field has size p^m for some prime p and integer $m \geq 1$.
- (b) **A field from polynomial remainders.** Let F be a field with q elements. Let $f \in F[X]$ have degree $m \geq 1$ and leading coefficient 1, and suppose that f is *irreducible*: it cannot be written as a product of two polynomials of positive degree. Consider the set

$$E = \{a_0 + a_1X + \cdots + a_{m-1}X^{m-1} : a_0, \dots, a_{m-1} \in F\}.$$

To add or multiply elements of E , first add or multiply the polynomials as usual, then take the remainder upon division by f . Show that these operations make E a field with q^m elements. *Hint*: Polynomial division and Bézout's identity work for polynomials over a field, just as they do for integers.

- (c) **An explicit example.** Show that $X^2 + X + 1$ is irreducible over $\mathbb{F}_2$. Apply part (b) to construct a four-element field. List its elements and compute the inverse of each nonzero element. Explain why $\{0, 1, 2, 3\}$ with integer arithmetic modulo 4 is not a field.
- (d) **Optional: existence for every prime power.** Let F be a finite field with q elements. We will show that irreducible polynomials over F exist in every positive degree, and then use part (b) to construct fields of every prime-power size.

- (i) Let $f \in F[X]$ be irreducible of degree $d_1 \geq 1$. Show that, for every integer $d_2 \geq 1$,

$$f \mid X^{q^{d_2}} - X \iff d_1 \mid d_2.$$

- (ii) Show that every irreducible factor of $X^{q^m} - X$ has multiplicity one, for every integer $m \geq 1$.
- (iii) Let N_d be the number of irreducible polynomials in $F[X]$ of degree d with leading coefficient 1. By counting degrees in the factorization of $X^{q^m} - X$, show that

$$q^m = \sum_{d \mid m} dN_d.$$

Use this identity and a simple upper bound on the contribution from proper divisors of m to prove that $N_m > 0$ for every $m \geq 1$. Finally, apply part (b) with $F = \mathbb{F}_p$ to conclude that a field with p^m elements exists for every prime p and integer $m \geq 1$.

- (e) **Optional: fields of the same size.** Let K and K' be fields with the same finite number of elements. Show that there is a bijection $\phi : K \rightarrow K'$ such that, for all $a, b \in K$,

$$\phi(a + b) = \phi(a) + \phi(b), \quad \phi(ab) = \phi(a)\phi(b), \quad \phi(1) = 1.$$

Such a map is called a *field isomorphism*. *Hint*: First show that every element of a field with q elements is a root of $X^q - X$.

Problem 5: A weak Johnson bound via a greedy procedure

Adapted from Homework 1, Problem 5 in Shashank Srivastava's *Topics in Coding Theory* course (Spring 2025).

Let $\mathcal{C} \subseteq \Sigma^n$ have relative distance at least δ , where $3/4 < \delta < 1$. We will give a greedy proof of a list-size bound at radius

$$\rho = 1 - 2\sqrt{1 - \delta}.$$

For a received word $g \in \Sigma^n$, write $\mathcal{L}(g, \rho) = \{h \in \mathcal{C} : \Delta(g, h) \leq \rho n\}$. Fix $0 < \varepsilon < 2\sqrt{1 - \delta}$ and consider the following procedure.

Greedy procedure

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 $S \leftarrow [n], \quad \mathcal{L}^* \leftarrow \emptyset$ 
while there is an  $f \in \mathcal{C}$  agreeing with  $g$  on at least  $\varepsilon n$  positions in  $S$  do
     $A(g, f) \leftarrow \{i \in S : g_i = f_i\}$ 
     $\mathcal{L}^* \leftarrow \mathcal{L}^* \cup \{f\}$ 
     $S \leftarrow S \setminus A(g, f)$ 
end while

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- (a) Show that the procedure runs for at most $1/\varepsilon$ iterations, so $|\mathcal{L}^*| \leq 1/\varepsilon$.
- (b) For every $h \in \mathcal{L}(g, \rho)$, show that some $f \in \mathcal{L}^*$ agrees with h on more than

$$\varepsilon(2\sqrt{1 - \delta} - \varepsilon)n$$

coordinates.

- (c) Choose ε appropriately and conclude that

$$\left| \mathcal{L}\left(g, 1 - 2\sqrt{1 - \delta}\right) \right| \leq \frac{1}{\sqrt{1 - \delta}}.$$

- (d) Explain why this procedure need not give an efficient list-decoding algorithm.